

● HARD

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

05

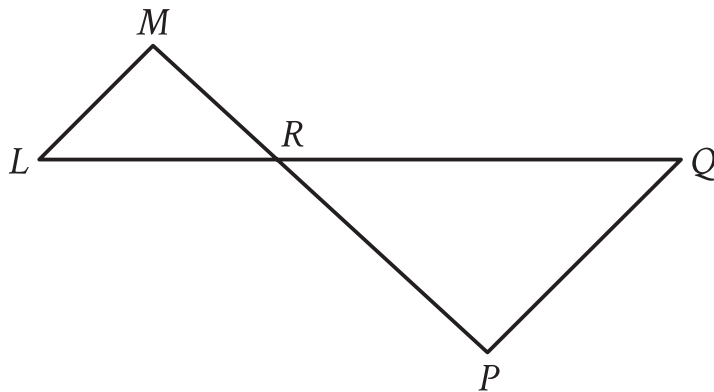
10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

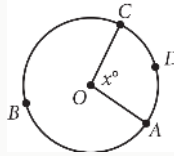
- Triangle upper L upper M upper R has a common vertex with triangle upper P upper Q upper R.
- The common vertex is upper R.
- A note indicates the figure is not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$, $\overline{LR}$, and $\overline{RP}$ are 6, 7, and 11, respectively. What is the length of $\overline{LQ}$?

- A. $\frac{119}{11}$
- B. $\frac{77}{6}$
- C. $\frac{113}{6}$
- D. $\frac{119}{6}$

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

- A. 18
- B. 27
- C. 36
- D. 81



The circle above has center O, the length of arc $\widehat{ADC}$ is 5π , and $x = 100$. What is the length of arc $\widehat{ABC}$?

- A. 9π
- B. 13π
- C. 18π
- D. $\frac{13}{2}\pi$

An isosceles right triangle has a perimeter of $94 + 94\sqrt{2}$ inches. What is the length, in inches, of one leg of this triangle?

- A. 47
- B. $47\sqrt{2}$
- C. 94
- D. $94\sqrt{2}$

Q5

GRID-IN

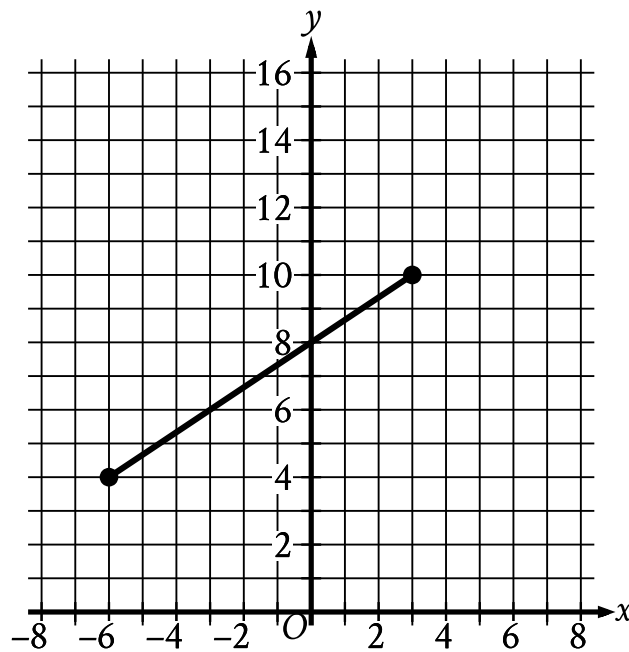
GEOMETRY AND TRIGONOMETRY

In triangle RST , angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 72 units, the length of $\overline{LK}$ is 24 units, and the area of triangle RST is 792 square units, what is the length of $\overline{KT}$, in units?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

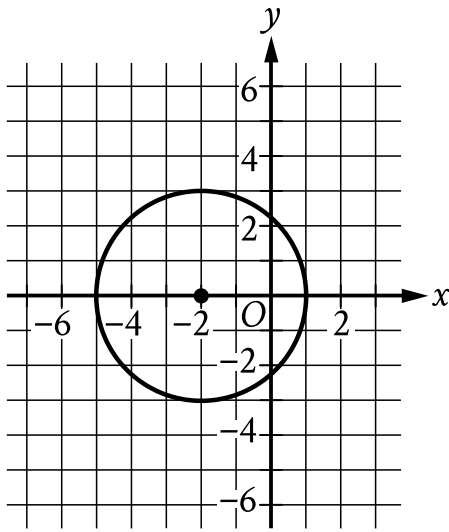
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The line segment slants gradually up from left to right.
- The line segment begins at the point (negative 6 comma 4).
- The line segment ends at the point (3 comma 10).

The line segment shown in the xy -plane represents one of the legs of a right triangle. The area of this triangle is $36\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

- A. 12
- B. 24
- C. $3\sqrt{13}$
- D. $18\sqrt{13}$



- The center of the circle is the point (negative 2 comma 0).
- Clockwise from top, the circle passes through the following points:
 - (negative 2 comma 3)
 - (1 comma 0)
 - (negative 2 comma negative 3)
 - (negative 5 comma 0)

Circle A (shown) is defined by the equation $x + 2^2 + y^2 = 9$. Circle B (not shown) is the result of shifting circle A down 6 units and increasing the radius so that the radius of circle B is 2 times the radius of circle A. Which equation defines circle B?

- A. $x + 2^2 + y + 6^2 = 49$
- B. $2x + 2^2 + 2y + 6^2 = 9$
- C. $x + 2^2 + y - 6^2 = 49$
- D. $2x + 2^2 + 2y - 6^2 = 9$

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles C and F are right angles. The length of $\bar{AB}$ is 2.9 times the length of $\bar{DE}$. If $\tan A = \frac{21}{20}$, what is the value of $\sin D$?

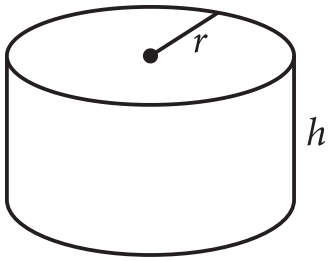
STUDENT-PRODUCED RESPONSE

○	/ ○	/ ○	○
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Quadrilaterals $PQRS$ and $WXYZ$ are similar, where P , Q , and R correspond to W , X , and Y , respectively. The measure of $\angle S$ is 135° , $PS = 45$, and $WZ = 9$. What is the measure of $\angle Z$?

- A. 5°
- B. 27°
- C. 45°
- D. 135°



The figure shown is a right circular cylinder with a radius of r and height of h . A second right circular cylinder (not shown) has a volume that is 392 times as large as the volume of the cylinder shown. Which of the following could represent the radius R , in terms of r , and the height H , in terms of h , of the second cylinder?

- A. $R = 8r$ and $H = 7h$
- B. $R = 8r$ and $H = 49h$
- C. $R = 7r$ and $H = 8h$
- D. $R = 49r$ and $H = 8h$